CASCADE, IOWA - November 3, 2026 - Source-linked research edition - Not adopted law

DRAFT - M-2 Heavy Industrial District Data Center and Performance Standards Amendment (November 3, 2026 revision)

> **DRAFT - FOR DISCUSSION ONLY - NOT ADOPTED LAW**
> Original prepared October 6, 2026; October 26 draft preserved. This **November 3, 2026 revised proposal** adds integrated definitions, clearer measurement and administrative cross-references, and Chicago-style legal notes. It is intended for City Attorney and Planning & Zoning review. This is a project-generated draft and is not legal advice. Section numbering, amendment mechanics, grandfathering, enforcement, and consistency with Iowa law should be reviewed by the City Attorney before introduction.
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> **Uniformity principle:** The ordinance is intentionally written so that zoning and performance standards apply throughout the M-2 Heavy Industrial District to all similarly situated businesses. Data centers are added as an expressly recognized use type, while noise, lighting, screening, water, emergency planning, generators, infrastructure, and other impact standards apply district-wide or through neutral, objective impact thresholds.

CITY OF CASCADE, IOWA
ORDINANCE NO. _____

AN ORDINANCE AMENDING CHAPTER 165 OF THE CASCADE ZONING ORDINANCE TO EXPRESSLY PERMIT DATA CENTERS AND HIGH-DENSITY COMPUTING FACILITIES IN THE M-2 HEAVY INDUSTRIAL DISTRICT AND TO ESTABLISH UNIFORM PERFORMANCE STANDARDS FOR M-2 INDUSTRIAL DEVELOPMENT

NOW, BE IT ENACTED by the City Council of the City of Cascade, Iowa, as follows:

> **Amendment mechanics / controlling baseline.** This proposal uses §165.11 in the City's scanned Chapter 165 (the M-2 district is on original PDF pages 45–49), as modified by the October 10, 2022 Ordinance 05-22 adding fitness businesses as M-2 permitted use B(18). **Existing A, B(1)–(18), C(1)–(4), D(1)–(7), E, F, and G(1) remain unchanged.** This review also recognizes the 2024 fence ordinance in §165.12(12); no local requirement in this draft overrides that ordinance without a separately enacted amendment. The 2009 zero-lot-line document within the original scan is marked **DRAFT**, so adoption cannot be assumed. The City Clerk/Attorney must confirm the complete amendment history and current codification before introduction. See the original M-2 scan and redlined comparison.

SECTION I. SECTION 165.11 M-2 - PERMITTED PRINCIPAL USES

Iowa Code authorizes zoning of industrial land use and requires district-wise uniformity for classes or kinds of buildings; this proposal does not confer project approval.[12]

Section 165.11, M-2 Heavy Industrial District, subsection B, is amended by adding the following permitted principal use:

19. Data Center or High-Density Computing Facility.

A building or group of buildings principally used to house computer servers, data-processing equipment, networking equipment, electronic storage equipment, or other high-density computing systems, including cloud computing, colocation, artificial-intelligence computing, scientific computing, cryptocurrency computing or mining, and substantially similar uses.

Minimum required off-street parking: one space per employee on the largest regularly scheduled shift, plus one space for each company vehicle regularly stationed at the premises.

Computer rooms, telecommunications equipment, and server equipment that are incidental and subordinate to another permitted principal use shall not constitute a separate Data Center or High-Density Computing Facility.

SECTION II. SECTION 165.11 M-2 - SPECIAL EXCEPTION USES

Section 165.11, M-2 Heavy Industrial District, subsection D, is amended by adding the following **proposed review classification**. The Board of Adjustment's statutory powers, including special exceptions and the separate standards governing variances, must be observed.[4]

8. Major Industrial Facility.

Any otherwise permitted principal use within the M-2 District meeting one or more of the following thresholds shall require Special Exception approval pursuant to Section 165.34(2):

1. Maximum anticipated coincident electrical demand of **five megawatts (5 MW) or greater**;
2. Anticipated industrial, process, cooling, or facility water demand of **25,000 gallons or more during any twenty-four-hour period**;
3. Aggregate stationary electrical generation capacity, excluding fire pumps, of **two megawatts (2 MW) or greater**;
4. Aggregate stationary battery or other electrical-energy-storage capacity of **five megawatt-hours (5 MWh) or greater**; or
5. A material expansion of an existing facility that causes the facility to exceed any threshold above.

The requirements of this subsection shall apply **without regard to the business type, technology, product, customer, tenant, ownership structure, or industrial process involved.** Similar physical impacts shall be treated consistently, and the City shall document the comprehensive-plan and public-welfare rationale for each objective threshold.[23]

SECTION III. SECTION 165.11 M-2 - UNIFORM PERFORMANCE STANDARDS

Existing §165.11(G)(1), “Supplementary District Regulations (Section 165.12),” remains unchanged. Add the following as §165.11(G)(2) through (19), immediately after the existing G(1). These provisions supplement, rather than replace, existing M-2 requirements and §165.12. Existing allowed uses, special exceptions, yard and height limits, signs, and later amendments remain in force.

2. Uniform Applicability

The requirements of this subsection apply uniformly to M-2 uses, structures, and operations unless a provision expressly establishes an objective physical or operational threshold. No requirement shall be imposed, waived, or varied solely because of the identity of the owner, operator, tenant, customer, industry, computing workload, product, or business model.

A lawful preexisting use, structure, or condition shall be treated in accordance with Sections 165.14 through 165.20 and other applicable law. To the maximum extent legally permissible, new construction, material expansion, replacement equipment, and materially modified operations after the effective date shall comply with the standards applicable to that work or modification.

3. Site Development Plan

Before issuance of a building permit for a new principal M-2 use or Major Industrial Facility, the applicant shall submit an engineered site development plan identifying, as applicable:

- buildings and structures;
- parking and loading areas;
- access roads;
- exterior mechanical and cooling equipment;
- generators;
- electrical substations and transformers;
- energy-storage systems;
- fuel and chemical storage;
- water and sewer connections;
- stormwater facilities;
- exterior lighting;
- landscaping and screening;
- fences;
- property boundaries; and
- existing sensitive receptors located within 1,000 feet of the property.

For a Major Industrial Facility, the plan shall additionally identify maximum anticipated electrical load, water demand, stationary generation capacity, fuel storage, cooling technology, principal noise-generating equipment, and proposed phasing.

For any Major Industrial Facility using stationary backup or on-site generation, the applicant shall provide a generator schedule identifying each engine's proposed manufacturer/model, rated electrical and thermal capacity, fuel type, exhaust configuration, fuel tank and containment, anticipated emergency and non-emergency operating modes, reasonably foreseeable simultaneous-operation scenarios, planned testing frequency and duration, and applicable state or federal emissions permits or permit-exemption determinations. Information requiring legitimate security protection may be submitted in an appropriately redacted public version and a protected version, only as authorized by Iowa law. The City shall not purport to issue or replace air-quality permits.

4. Operational Noise

All M-2 uses shall be designed, constructed, operated, and maintained so that noise attributable to the use does not exceed the following standards, measured in accordance with generally accepted professional acoustic practice:

a. M-2 property-line standard: 65 dBA and 75 dBC hourly average at any property boundary, except where the residential/sensitive-receptor standard below is more restrictive.

b. Residential or sensitive-receptor standard: At each existing dwelling, school, licensed child-care facility, hospital, nursing facility, place of worship, public library, or comparable occupied sensitive use, compliance shall be evaluated **inside the closest normally occupied room on the side of the structure nearest the M-2 noise source**. The initial assessment shall include the nearest existing qualifying occupied structure. Additional qualifying structures shall be evaluated where source orientation, terrain, or shielding could produce higher exposure.

The sound-level meter microphone shall be placed **within five (5) feet of the interior face of the nearest source-facing exterior wall**, but **at least 1.5 feet (0.5 meter) from that wall and other large reflecting surfaces**, where physically practicable, at approximately four (4) to five (5) feet above the finished floor. **All doors and windows shall be closed throughout the measurement. In a dwelling with a controllable forced-air HVAC blower, the fan control shall be set to ON for continuous operation during the entire indoor measurement, with the thermostat set within 65°F–75°F where reasonably practicable; the detailed protocol in subsection G(5) applies.** The qualified acoustical professional shall identify the room and measurement positions and document any necessary deviation from these placement requirements. The following **indoor, facility-attributable one-hour equivalent continuous sound levels (Leq,1h)** shall not be exceeded:

- 55 dBA and 65 dBC from 7:00 a.m. to 10:00 p.m.; and
- 50 dBA and 60 dBC from 10:00 p.m. to 7:00 a.m.

c. Tonal noise: Where an industrial sound source produces a prominent recurring tone, whine, screech, or hum, the applicable A-weighted limit shall be reduced by five (5) decibels.

d. Measurement: The one-hour average is the energy-equivalent continuous sound level (Leq,1h), not an arithmetic mean of decibel readings. A qualified acoustical professional shall use a calibrated Type 1 or Type 2 sound-level meter appropriate for the purpose, identify A- or C-weighting, document meteorology for exterior measurements, receiving location, background sources, room conditions for interior measurements, and source operating conditions. Outdoor source assessment shall use a documented ANSI/ASA S12.9 or successor environmental-noise assessment method. Indoor sensitive-receptor assessment shall use a documented method appropriate for room sound and low-frequency industrial noise, including ANSI/ASA S12.72 and S12.9 Part 7 where applicable.[1718] The City shall evaluate only facility-attributable noise using a documented background-noise adjustment procedure, accounting for unrelated indoor sound sources. Entry into an occupied structure for testing requires occupant consent or other lawful authority. If access is unavailable, the professional may report modeled indoor exposure with explicit assumptions, but shall not represent an outdoor measurement as a direct indoor compliance test. These measurement protocols shall not be construed to set a new statewide noise standard.[910]

e. Voluntary indoor access and outdoor alternative: The owner or lawful occupant of a sensitive-receptor structure may decline voluntary indoor noise testing. Declining consent does not invalidate a complaint or waive other remedies authorized by law. Interior or exterior entry onto private property requires consent or an independent lawful basis.

If indoor testing is declined, the qualified acoustical professional shall measure the facility-attributable Leq,1h outdoors at the point on the sensitive-receptor parcel boundary closest to the industrial property, provided the location is lawfully accessible and acoustically representative. Otherwise, a nearby lawfully accessible representative location shall be selected and the departure documented. The report shall document background conditions, microphone position, source operating conditions and any distance, shielding or reflection adjustments needed to estimate sound incident on the protected structure.

To assess the proposed indoor thresholds, the professional shall use documented, source-spectrum-appropriate outdoor-to-indoor noise reduction (OINR) or equivalent acoustical transmission-loss data for the building assembly, including walls, windows, doors and openings. Separate A- and C-weighted estimates and low-frequency assessment shall be provided where applicable. A calculated indoor result shall be labeled an **estimate**, not a direct indoor measurement. The applicable M-2 property-line limit remains unchanged.

Exterior-wall insulation R-values in Iowa energy requirements measure thermal resistance, not acoustical isolation. R-values may inform assembly identification but shall not be converted directly into dBA/dBC corrections. If reliable facade sound-reduction evidence is unavailable, a zero-decibel reduction may be used for cautious preliminary screening only, not by itself as conclusive proof of an indoor violation. The source of any acoustic correction and its uncertainty shall be disclosed.[2123]

5. Acoustic Study and Verification

Every Major Industrial Facility shall provide, at the applicant's expense:

1. A **preconstruction baseline acoustic evaluation**, completed before site construction or grading materially changes ambient sound, documenting representative exterior A- and C-weighted hourly equivalent levels (Leq,1h) at nearby sensitive-receptor parcels and M-2 property boundaries. The evaluation shall include **voluntary indoor baseline measurements** inside nearby existing sensitive-receptor structures wherever the property owner or lawful occupant consents, as specified below;
2. A predictive acoustic model demonstrating compliance at property boundaries and nearby sensitive receptors under maximum reasonably anticipated operating conditions, including hot-weather conditions where relevant;
3. A post-construction compliance evaluation within sixty (60) days after commencing normal operations at reasonably representative load and weather conditions, or within a documented alternative period approved for commissioning constraints; and
4. A recurring compliance evaluation at least once every three (3) years, or following a material modification reasonably expected to affect noise levels.

Preconstruction indoor-receptor baseline and consent. The applicant shall identify the closest existing dwelling or other sensitive-receptor structures and any additional receptors reasonably expected to receive greater facility-related noise because of equipment orientation, terrain, or shielding. Before construction, the applicant shall offer the owner or lawful occupant of each selected structure an opportunity, **at no cost to that owner or occupant**, to consent to baseline testing within the nearest normally occupied room facing the proposed industrial source. When consent is granted, a qualified acoustical professional shall record representative **daytime and nighttime indoor dBA and dBC Leq,1h levels** under the placement, closed-door, and closed-window conditions described in subsection G(4)(b), together with operating indoor HVAC or ventilation conditions, existing sound sources, weather where relevant, and measurement times. The

study should obtain paired exterior measurements at the same structure when lawful access and conditions permit. This provides a documented **preconstruction ambient baseline**, not a presumption that existing sound is attributable to the proposed facility.

Declined or unavailable access. Neither a property owner nor a lawful occupant shall be required by this subsection to permit voluntary entry. The applicant shall document reasonable efforts to offer indoor baseline testing and record whether consent was granted or declined without disclosing unnecessary private household information. Where indoor consent is declined or lawful access is unavailable, the preconstruction study shall perform the exterior alternative in subsection G(4)(e), including a source-facing receptor-parcel boundary measurement **only where lawful access is available**, or the nearest suitable lawful exterior location. Any estimated interior baseline shall use independently substantiated acoustical transmission-loss assumptions; **Iowa exterior-wall thermal R-values shall not be converted into decibel adjustments**. Estimated indoor levels shall be identified as *modeled*, not as measured values. A refusal of voluntary interior testing shall not be treated as a waiver of the applicable noise protections.

Thermostat and HVAC fan setting during indoor testing. During every indoor preconstruction baseline, postconstruction verification, and complaint-response acoustic measurement, **the forced-air HVAC circulation fan shall be set to ON for continuous operation throughout the entire measurement interval**, wherever the building has an occupant-controllable fan ON setting. **AUTO is not the default for this test.** For dwellings, the heating/cooling thermostat setpoint shall be within **65°F to 75°F (inclusive)** where reasonably practicable and consistent with occupant health and safety. Heating and cooling demand may still cycle normally in response to the thermostat; the blower must remain on between those calls. Do not deliberately force heating or cooling cycles. The qualified acoustic professional shall verify and record the actual continuous blower operation, fan speed or mode, system type, thermostat setpoint and mode, measured interior temperature, heating/cooling cycles, other ventilation, and any deviation. These identical test settings shall be used for the comparable preconstruction and postconstruction measurements, or differences must be explained and accounted for in the technical assessment. Continuous fan sound is part of the measured indoor background and may mask industrial noise; the report shall distinguish facility-attributable sound from HVAC sound using a technically appropriate method. Supplemental measurements under other fan settings may inform the assessment but do not replace the required fan-ON measurement. If no blower or continuous-ON control is present, or use would be unsafe or medically inappropriate, document the reason and measure under representative existing conditions; lack of such equipment does not invalidate a complaint. The **65–75°F setpoint and mandatory fan-ON protocol are proposed local measurement conditions**, not Iowa residential code requirements.[25]

Evidence for a claimed increase after construction. A finding that a Major Industrial Facility **increased noise inside a neighboring sensitive-receptor structure** shall be supported by reliable evidence of the **preconstruction indoor baseline** and a comparable **postconstruction indoor level**, measured in the same room when practicable, **with exterior doors and windows closed**, using the same A- and C-weighted Leq,1h metrics and comparable daytime/nighttime periods, ventilation settings, and sound-source conditions. The analysis must establish that the later level is higher than the earlier level beyond the stated measurement uncertainty and must separately substantiate the amount **attributable to the facility**, rather than assuming all ambient sound comes from that facility. A preconstruction home need not have been silent; the relevant question is whether its indoor noise demonstrably increased because of the facility.[24]

Where an owner declined the preconstruction indoor evaluation, the City may consider credible earlier sound surveys, contemporaneous records, and the documented exterior-to-interior acoustic model described in subsection G(4)(e). An unsupported assertion of an increase alone shall not

establish a violation **based solely on a change from preconstruction conditions**. This evidentiary requirement **does not prevent anyone from submitting a complaint**, does not require a historical baseline to investigate or enforce an **absolute indoor or M-2 property-line limit**, and does not displace rights or remedies under Iowa nuisance law.[924]

The required acoustic model and post-startup testing shall examine ordinary peak operations, reasonably foreseeable simultaneous generator testing or emergency backup operation, identifiable low-frequency or tonal equipment sources, and nearby residential and other sensitive receptors. Where safety or operational requirements prevent actual simultaneous generator testing, a qualified professional may combine measured source data with documented predictive modeling, with assumptions stated in the report.

Any separate structure-borne or ground-vibration concern must be evaluated with measurements appropriate to vibration (such as peak particle velocity), not inferred from dBA/dBC sound measurements. A numeric vibration ceiling is **not adopted in this draft** pending independent engineering and legal review.

The City may require reasonable corrective measures where testing demonstrates noncompliance.

6. Generator and Elevated-Noise Testing

Routine and pre-planned testing of stationary generators or other equipment producing substantially elevated noise shall occur only between **8:00 a.m. and 6:00 p.m., Monday through Friday**, except where operation outside those hours is:

1. Required during an actual emergency;
2. Necessary to protect life, property, or critical equipment;
3. Required by a governmental agency, applicable code, manufacturer, utility, insurer, or reliability authority; or
4. Specifically authorized by the Zoning Administrator because the testing cannot reasonably be completed during ordinary testing hours.

This restriction applies uniformly to M-2 businesses operating such equipment. "Emergency generator" as classified for Iowa air-quality permitting follows applicable state air regulations; local zoning testing-hour rules do not alter air-permit classifications or authorize impermissible emissions.[8]

7. Generator Operating Records

Any M-2 facility having aggregate stationary generation capacity of two (2) MW or more shall maintain records identifying:

- date and duration of each operation;
- whether the operation was emergency or non-emergency;
- reason for operation; and
- equipment operated.

Records shall be retained for at least three (3) years and made available to the City upon reasonable request for zoning-compliance purposes, subject to applicable public-record and confidentiality law.

8. Major Industrial Facility Setbacks

Existing M-2 dimensional requirements shall remain applicable generally. In addition, for a new Major Industrial Facility:

1. A principal building shall be located at least **100 feet from a property boundary adjoining a residential zoning district**.
2. Stationary generators, exterior cooling equipment, large transformers, substations, compressors, industrial fans, and comparable exterior mechanical equipment shall be located at least **200 feet from a residential zoning district boundary**.
3. The Board of Adjustment may approve a reduced equipment setback, but never less than 100 feet, where an applicant-funded independent acoustic analysis demonstrates that all applicable noise standards will be met under maximum reasonably anticipated operating conditions.

Nothing in this subsection authorizes a setback reduction that conflicts with any other applicable code, easement, fire requirement, or utility clearance.

9. Exterior Lighting

All new or replacement exterior lighting within the M-2 District shall:

1. Be fully shielded and directed downward;
2. Be designed to prevent direct glare onto residential property and public streets;
3. Not exceed **0.5 foot-candle at a residential zoning boundary**, excluding existing public street lighting; and
4. Use automatic controls or reduced lighting levels during periods when full illumination is unnecessary, except for safety, security, and emergency lighting.

10. Screening

Exterior mechanical equipment, generators, fuel tanks, refuse areas, substations, and comparable industrial equipment visible from an adjoining residential district or public right-of-way shall be screened through landscaping, berms, walls, fencing, equipment enclosures, or a combination thereof.

Screening shall not interfere with required ventilation, fire access, utility clearance, security, or equipment safety.

11. Water Use and Cooling

Any M-2 use reasonably expected to use **25,000 gallons or more of water in a twenty-four-hour period** shall provide:

1. Projected average and peak water demand;
2. Identification of the proposed water source;
3. Applicable Iowa Department of Natural Resources permits, approvals, or documentation, with any determination of a **regulated** or **nonregulated** water use made under Iowa Code §455B.261 and applicable implementing rules, not inferred from the City's impact-review threshold alone;[7]
4. An engineering analysis demonstrating adequate water supply without unreasonable interference with existing users;
5. Separate metering of material industrial or cooling-water use where practicable; and

6. Annual water-use reporting to the City.

Once-through cooling using potable municipal water is prohibited for new Major Industrial Facilities.

Closed-loop or non-evaporative cooling shall be preferred. An evaporative system may be approved where professional engineering demonstrates that the alternative will not create an unreasonable demand upon municipal water resources and adequately controls drift, plume, condensation, and icing.

12. Electrical Utility Verification

Before issuance of a building permit for any M-2 use having maximum anticipated electrical demand of five (5) MW or more, the applicant shall provide written documentation from the serving electric utility confirming that:

1. The proposed maximum electrical demand has been disclosed;
2. Required utility infrastructure and service upgrades have been identified sufficiently for the utility to provide service;
3. Responsibility for project-specific utility improvements has been addressed between the utility and customer; and
4. The facility's electrical service may be constructed and operated in accordance with applicable utility, electrical, and safety requirements.

Nothing in this ordinance regulates electric rates, utility service territories, generation procurement, or matters preempted or otherwise committed by law to a public utility, electric cooperative, state agency, or federal authority.

13. Fuel, Batteries, Hazardous Materials, and Emergency Response

A Major Industrial Facility shall provide an Emergency Response Plan to the City and appropriate emergency-response agencies before occupancy.

The plan shall identify, where applicable:

- emergency contacts;
- facility access;
- electrical disconnects;
- generator locations;
- fuel quantities and shutoffs;
- coolant and chemical inventories;
- battery-energy-storage systems;
- fire-suppression systems;
- evacuation procedures;
- spill response; and
- procedures for providing emergency responders access to the site.

The plan shall be reviewed following material facility modifications and at reasonable intervals requested by the City or responding agencies.

14. Heat Rejection, Drift, Condensation, and Icing

No M-2 mechanical exhaust, cooling, or heat-rejection system shall be operated in a manner that causes ice, condensate, cooling-tower drift, mist, or similar discharge to accumulate upon an adjoining property or public right-of-way or create an unreasonable safety hazard.

15. Stormwater and Drainage

All new M-2 development shall comply with applicable City, state, and federal stormwater requirements.

A Major Industrial Facility shall submit engineered stormwater calculations demonstrating that development will not cause an unlawful or unreasonable increase in off-site flooding, erosion, or drainage impacts, **subject to Iowa Code §364.3 limitations on municipal runoff and topsoil requirements**. An independently imposed more restrictive runoff standard may require the City to bear specified engineering, construction, or property costs, and shall not be presumed authorized by this zoning section.[12]

The property owner shall be responsible for repair of documented project-caused damage to City drainage infrastructure to the extent permitted by law.

16. Public Infrastructure

Where an M-2 development creates a documented need for road, water, sewer, drainage, emergency-service, or other public improvements beyond existing system capacity, the City may require a development or infrastructure agreement allocating proportionate project-related costs to the applicant to the extent permitted by Iowa law.

This provision shall apply equally to every M-2 development creating the same type and degree of infrastructure burden.

Where substantial construction hauling is reasonably expected to affect City roads, the City may, pursuant to an independently authorized permit or written road-use agreement, require a preconstruction photographic/engineering condition survey, defined haul routes and axle/load limits, reasonable maintenance and restoration terms, postconstruction verification, and proportionate financial security supported by an engineering estimate. No fixed multiplier or new fee is prescribed here without legal review.

17. Decommissioning of Major Industrial Facilities

An application for a Major Industrial Facility shall include a decommissioning plan describing:

1. Anticipated facility life;
2. Equipment and structures to be removed;
3. Environmental remediation and site restoration;
4. Estimated decommissioning cost;
5. Salvage assumptions; and
6. Proposed method of financial assurance.

A facility shall be considered abandoned when its principal industrial operation has ceased continuously for twelve (12) months, excluding temporary shutdowns caused by reconstruction, documented market conditions accompanied by evidence of a bona fide intent to resume operations,

casualty, litigation, government order, utility interruption, or similar circumstances approved by the Zoning Administrator.

Following abandonment, decommissioning shall be completed within twelve (12) months unless extended for good cause.

The form and amount of financial assurance shall be established in a manner approved by the City Attorney and shall be reasonably related to the documented cost of decommissioning and site restoration.

A qualified professional's decommissioning cost estimate shall be reviewed at least once every three (3) years and following a material expansion or ownership/operator change that materially alters restoration obligations. The City may require adjustment of assurance to the reasonable documented cost of performance, including contingencies, only through a legally authorized and objectively administered instrument.

To the extent enforceable under Iowa law, approval conditions and any separate recorded agreement shall identify responsible owners and operators, notice requirements upon transfer, acceptable successor security, and procedures for default and lawful recovery. Private contracts shall not transfer away obligations imposed by law or approval conditions.

18. Material Expansion

An approved Major Industrial Facility shall require an updated Special Exception review where a proposed modification would increase any of the following by **twenty percent (20%) or more** above the previously approved maximum:

- electrical demand;
- water demand;
- stationary generation capacity;
- energy-storage capacity;
- exterior mechanical-equipment capacity; or
- developed building area.

Crossing a Major Industrial Facility threshold for the first time shall require Special Exception approval regardless of percentage increase.

19. Compliance, Inspection, and Corrective Action

The Zoning Administrator may require documentation reasonably necessary to verify compliance with conditions of approval and the objective standards of this section. Where specialized testing or engineering is reasonably required because of a credible compliance question involving a Major Industrial Facility, the City may require the owner or operator to provide qualified professional testing, subject to applicable law and any fee or cost-recovery provisions adopted by the City.

Each Major Industrial Facility shall submit a brief annual compliance certification to the Zoning Administrator by March 1 for the prior calendar year, identifying the facility owner and operator; material changes in electricity demand, generation equipment, cooling and water use; whether prescribed acoustic tests and generator logs have been maintained; reported material incidents or violations; and the status of financial assurance. The certification is a compliance statement, not automatic approval of unverified representations.

Where reasonably necessary and authorized by an adopted fee schedule or agreement, the City may retain or approve an independent qualified professional to review applicant studies and required remedial testing at reasonable, documented applicant cost. For complex matters, the City may require a deposit or escrow established through lawful procedures with an accounting and refund of unused amounts. The applicant must not control or direct the independent expert's findings.

Nothing in this section limits other remedies available under Chapter 165 or applicable law.

SECTION IV. SECTION 165.41 - DEFINITIONS

Section 165.41 is amended by adding the following definitions, preserving existing Chapter 165 definitions unless expressly changed. For **City zoning classification and enforcement**, the following local definitions govern. Iowa's tax-law definition of "data center" is an interpretive comparator, not an automatic incorporation. State building, air, water, waste, and stormwater definitions control only where their respective legal programs apply. A separately adopted state or federal permit remains required when applicable.[5][6][7][8][11][12]

The current 86-term statutory, Oxford, Merriam-Webster, and legal-dictionary comparison is an **interpretive research appendix**. Its general-language summaries do not override the enacted text of this ordinance or the controlling statute. A standalone companion draft for Section 165.41 is also maintained in `proposals/DRAFT-ORDINANCE-CH165-DEFINITIONS-2026-11-03.md`.

Data Center or High-Density Computing Facility: A building, structure, or portion thereof principally used for networked servers, electronic data processing, digital storage, artificial-intelligence, cloud, colocation, scientific, or cryptocurrency computing. Incidental computer rooms and telecommunications equipment subordinate to another principal use are excluded.

Major Industrial Facility: An M-2 use exceeding any one threshold in proposed §165.11(D)(8): 5 MW coincident demand, 25,000 gallons in any 24 hours, 2 MW aggregate stationary generation, or 5 MWh storage. Also includes a modification that crosses a threshold.

Material Expansion: A modification increasing an approved maximum by 20 percent or more for electrical demand, water demand, stationary generation, energy storage, exterior mechanical capacity, or developed building area, or crossing a Major Industrial Facility threshold for the first time.

Sensitive Receptor: An occupied dwelling, school, licensed child-care facility, hospital, nursing facility, place of worship, public library, public park facility, or comparable occupied use requiring protection from continuous industrial noise.

Ambient Noise / Baseline Sound: The sound environment at a designated receiving location without the regulated facility's incremental contribution, determined under representative day, night, and meteorological conditions. Ambient noise is not the Iowa air-quality term "ambient air."[8][10]

Facility-Attributable Noise: The sound contribution attributable to a regulated M-2 use at the specified receiving point, including reasonably foreseeable simultaneous operation of equipment; exclude unrelated transportation, natural, and third-party background noise through an approved, documented acoustical procedure.[10]

dBA: A-weighted decibels of sound pressure, measured with calibrated equipment using the applicable professional noise measurement method.

dBC: C-weighted decibels of sound pressure, which place greater relative weight on low-frequency sound than A-weighting.

Hourly Average Sound Level (Leq,1h): The energy-equivalent continuous sound-pressure level over sixty (60) minutes, expressed using the specified A- or C-weighting; intermittent contributions are averaged on an acoustic energy basis, not by taking an arithmetic mean of decibel values. The qualified professional shall identify source operation, background sound, measurement location, and calibration.[10]

Prominent Tone / Tonal Noise: An identifiable recurring or steady audible frequency component distinguishable from neighboring sound bands when evaluated under the ordinance's specified professional acoustical method.

Acoustic Study: A qualified professional's documented baseline survey, maximum-design-operation model, receptor analysis, assumptions, equipment schedule, and verification protocol.

Maximum Anticipated Coincident Electrical Demand: The greatest reasonably anticipated simultaneous electrical load, expressed in MW, for the complete site during normal or design-maximum operation. A sum of independent equipment nameplate ratings does not establish coincident site demand without a documented load calculation.

Aggregate Stationary Generation Capacity: The sum of rated electrical outputs of all stationary generators covered by the section, in megawatts, excluding fire pumps only where expressly provided.

Aggregate Electrical-Energy-Storage Capacity: The summed rated stored electrical energy of covered stationary storage systems, in megawatt-hours, using documented nameplate energy ratings.

Exterior Mechanical Equipment: Permanently located outdoor fans, compressors, cooling equipment, generators, transformers, heat-rejection systems, and comparable mechanical installations serving an M-2 principal use.

Property Boundary / Lot Line: The legally established boundary of the parcel under review as supported by a recorded plat, survey, deed, or other legally controlling boundary record.

Residential Zoning District Boundary: The official delineation between an M-2 parcel or district and land designated residential by the City's controlling zoning map.

Sensitive-Receptor Measurement Location: The specified exterior wall, occupied outdoor-use area, or other objectively identified receiving point at a sensitive receptor, documented on an approved site/acoustic plan.

Once-Through Cooling: A process in which municipal potable water passes through a cooling system once and is discharged rather than being recirculated for heat rejection.

Closed-Loop Cooling: A recirculating coolant system intended to keep working fluid contained, subject to any limited makeup or process losses specifically disclosed.

Emergency Response Plan: A written, site-specific plan documenting access, contacts, utility isolation, fuel, chemicals, battery systems, suppression, evacuation, spills, and coordination with first responders.

Decommissioning: The safe disconnection, dismantling, lawful removal, cleanup, and site restoration work required at the end of a covered facility's life.

Abandonment: Continuous cessation of a facility's principal approved industrial operation for the period specified in §165.11(G)(17), subject to its listed reasonable exceptions.

Financial Assurance: A surety, letter of credit, escrow, performance bond, trust, or other legally authorized and enforceable arrangement in a reasonable amount covering documented decommissioning and site restoration duties. The statutory definition applicable to sanitary-disposal facilities under Iowa Code §455B.301(9) does not itself grant the City authority to impose that requirement on M-2 data centers; a separate legal basis must be confirmed by the City Attorney.[13]

Independent Qualified Professional: An appropriately licensed or credentialed expert with relevant technical competence and no financial or managerial interest that would compromise independent evaluation.

Material Modification (for monitoring): A change in equipment, site layout, generation operation, or other covered operation reasonably likely to change the regulated environmental or infrastructure impact.

Relationship to other Iowa-defined terms

For interpretation **within the respective state-regulated subject**, the following meanings are governed by the stated Iowa authority: water-allocation terms "aquifer," "beneficial use," "nonregulated use," and "regulated use" (Iowa Code §455B.261); state air-emissions terms "air contaminant," "emission," and "major stationary source" (§455B.131 and Iowa Administrative Code 567—22.1); hazardous-condition response terms (§455B.381); hazardous-waste terms (§455B.411); and stormwater permitting terms (Iowa Administrative Code 567—60.1 and incorporated federal definitions). Those terms **are not silently repurposed** as independent municipal zoning thresholds. State-defined "major stationary source" is distinct from the City's proposed "Major Industrial Facility."[781114]

Interpretation and conflicts

The originally scanned M-2 dimensional standards are a 25-foot front yard, 25-foot rear yard, 20-foot side yard, 20-foot street-side yard for corner lots, and 60-foot maximum height, except where duly adopted later provisions modify them. The source PDF itself includes an unsigned and undated **2009 draft** proposing a zero-lot-line exception; the scan alone cannot establish whether it was enacted. A later Ordinance 05-22 adds permitted fitness uses as item B(18), and Ordinance 19-24 permits fences up to eight feet in M-2 subject to its additional conditions. **No unconditional assertion of current-law completeness is made.**[1516]

The proposal preserves existing Cascade definitions of "Building/Structure," "Accessory Use," "Lot," and "Dwelling" except where a later duly adopted amendment controls. The City Attorney shall verify all intervening amendments to the 1992 base zoning code and confirm final section numbering, cross-references, and special-exception authority before introduction.[46]

SECTION V. CONFLICTS AND MORE RESTRICTIVE REQUIREMENTS

Nothing in this ordinance relieves a property owner or operator from compliance with any other applicable City, county, state, federal, utility, fire, building, environmental, or safety requirement. Where two applicable zoning provisions regulate the same subject and can lawfully be applied together, the more restrictive provision shall control, consistent with Section 165.03.

SECTION VI. SEVERABILITY

If any section, subsection, paragraph, sentence, clause, phrase, or application of this ordinance is adjudged invalid or unconstitutional, such adjudication shall not affect the validity of the ordinance as a whole or any other section, provision, or application not adjudged invalid or unconstitutional.

SECTION VII. EFFECTIVE DATE

This ordinance shall be in full force and effect after its final passage, approval, and publication as provided by law.

Chicago Notes and Bibliography - source notes

Superscript note markers in this proposal point to these numbered sources. Notes are documentary authorities and **are not operative zoning terms**. Project-specific engineering assumptions and proposed numerical thresholds still require independent validation.

NON-CODIFIED DRAFTING NOTES AND SOURCE BASIS

> The material below is explanatory and is **not intended to be codified as part of Chapter 165**. It documents why specific concepts appear in the draft and identifies issues for legal and technical review.

A. October 1, 2026 Simple Mining Community Forum

The October 1 forum materially clarified the proposed Cascade project. The company represented that its current concept is approximately 60,000 square feet on roughly 15.2 acres, with an approximately 40 MW electrical load, 24/7/365 operation, closed-loop hydrocooling with dry/air-cooled heat rejection, approximately 15 backup diesel generators, and 20-25 projected permanent jobs. The company also expressed willingness in principle to accept enforceable noise, water, emergency-planning, screening, decommissioning, and expansion-related conditions.

These statements are representations from the forum and are not substitutes for final engineered plans, permits, utility agreements, or binding conditions. Repository record: `dataset/updates/28_SIMPLE_MINING_COMMUNITY_FORUM_2026-10-01.md`.

B. Uniform District-Wide Approach

This draft intentionally avoids regulating Simple Mining, cryptocurrency mining, or data centers as uniquely disfavored businesses. It adds data centers as an expressly recognized M-2 use while regulating physical impacts through standards that apply throughout the district or through neutral thresholds such as MW demand, water use, generator capacity, and battery-storage capacity.

The approach should be reviewed for consistency with Iowa Code Chapter 414, including the zoning-uniformity provisions of Iowa Code section 414.2.

Official Iowa Legislature source:
https://www.legis.iowa.gov/docs/code/414.2.pdf

C. Nearby Iowa Comparators

Linn County

Linn County's 2026 data-center ordinance provides useful measurable provisions involving setbacks, acoustic studies, recurring sound testing, emergency planning, road impacts, and water review. Its noise framework includes 55 dBA / 65 dBC for adjacent residential occupied structures and 65 dBA / 75 dBC for adjacent nonresidential occupied structures, along with recurring acoustic testing.

Official ordinance:
https://www.linncountyiowa.gov/DocumentCenter/View/27695/PA26-0001-Ordinance-PDF

Staff report:
https://www.linncountyiowa.gov/DocumentCenter/View/27694/PA26-0001-Data-Centers-Ordinance-Staff-Report---3rd-Consideration-PDF

Jones County

Jones County Ordinance 2025-02 includes special-use/site-plan review, applicant-funded third-party noise analysis, cooling and water documentation, electric-utility verification, emergency planning, and decommissioning. It includes continuous-noise limits and a five-decibel adjustment for prominent steady tones, and requires liquid/closed-loop cooling for the regulated use.

Copy hosted by Linn County:
https://www.linncountyiowa.gov/DocumentCenter/View/27656/Jones-County-Data-Center-Ordinance

Other comparators

The project repository also tracks the City of Palo EU-1 district, Jackson County working drafts, Allamakee County policy, Norwalk Project West, and West Des Moines data-center development examples in `dataset/policy/06_IOWA_DATA_CENTER_POLICY_COMPARATORS.md`.

D. Water Threshold

The 25,000-gallon-per-day threshold is used as an objective industrial-impact trigger because Iowa water-use permitting law generally uses a 25,000-gallon quantity in a 24-hour period as a significant regulatory threshold. City Attorney and engineering review should confirm the best local wording and interaction with Iowa DNR permitting.

Iowa Administrative Code Chapter 50:
https://www.legis.iowa.gov/docs/iac/chapter/567.50.pdf

E. Acoustic Standards

The draft's dBA/dBC structure is intended to address both ordinary audible sound and low-frequency industrial hum. The reference to ANSI/ASA S12.9 provides a recognized environmental-noise measurement framework rather than creating a project-specific testing method.

November 3 indoor-receptor revision: Proposed §165.11(G)(4)(b) now evaluates the nearest normally occupied interior room, within five feet of the source-facing exterior wall, with doors and windows closed. The separate M-2 property-line standard remains outdoors. The five-foot maximum placement distance is a **proposed local rule**, not a distance mandated by ISO or ANSI. A 0.5-meter minimum clearance from reflective surfaces and documenting room conditions draw on accepted acoustic measurement principles.[18]

Substantive policy question: The numerical 55/65 dBA/dBC daytime and 50/60 dBA/dBC nighttime limits were carried forward unchanged, but an *indoor* limit is not equivalent to the original *outdoor* limit because walls and closed windows attenuate exterior sound. WHO's community-noise guidance identifies 30 dBA Leq for bedrooms at night as a sleep-protection guideline. The proposed 50 dBA nighttime indoor limit should therefore be explicitly reviewed by an independent acoustician and the City Attorney, not presented as a WHO-derived or state-mandated standard.[19] Closing windows also removes the draft's prior direct protection of outdoor living areas. Occupant consent or other lawful authority is required for indoor access; modeled exposure is a screening tool, not a substitute for a direct measurement.

Iowa home envelopes after 2015 (research, not a codified noise reduction credit): Iowa's statewide residential energy code remained the **2012 IECC with state amendments** following its April 1, 2014 effective date. It requires thermal insulation and a blower-door air-sealing result not exceeding **4 ACH50**, and allows common wood-frame prescriptive exterior wall constructions such as **R-20 cavity insulation or R-13 cavity plus R-5 continuous insulation**. These are *energy-efficiency* provisions, **not** requirements that every residence attenuate noise by a fixed number of dBA or dBC.[20]

FHWA's *Noise Measurement Handbook*, Table 6-1, illustrates **20 dB outside-to-inside reduction** for light-frame buildings with ordinary **closed** windows, **25 dB** for light-frame/storm-window assemblies and **25–35 dB** for listed masonry/window combinations. The values are **traffic-noise modeling assumptions**, **not Iowa housing acoustic standards**. Applying 20 dB to a hypothetical 65 dBA exterior value yields about 45 dBA inside **only if that particular sound spectrum and building actually achieve 20 dB attenuation**. These figures cannot be used as a blanket **20 dBC** deduction for low-frequency data-center, transformer or HVAC hum.[21]

For local context, Cascade Code **§6-12-8**, Ordinance **02-22** (August 22, 2022), adopted the current State of Iowa Building Code by reference. Dubuque County was identified as IECC Zone **5A**, Delaware County as **6A**, under the 2012-era county map; neither county classification establishes acoustic transmission loss. The City and acoustician should **not presume a specific residential noise attenuation solely from a construction date**, and should identify facade-specific, frequency-dependent performance or clearly labeled modeling assumptions.[22] The full building-code and exterior-noise review is part of the repository's non-codified evidence, not a new enforceable provision.

ANSI/ASA standards catalog:
https://webstore.ansi.org/

F. Stationary Generators

The draft distinguishes routine testing from actual emergencies and requires uniform testing-hour rules across M-2 businesses. Federal stationary-engine requirements should continue to control where applicable; zoning should not attempt to replace air permitting or emissions regulation.

U.S. EPA stationary-engine information:
https://www.epa.gov/stationary-engines

G. Items Better Addressed Outside the Zoning Ordinance

The following matters are generally better addressed through a development agreement, economic-development agreement, utility agreement, or other lawful contractual mechanism rather than being written as project-specific zoning requirements:

- LEED certification or equivalent sustainability certification;
- local hiring and workforce-development commitments;
- wage or benefit commitments;
- community-benefit payments;
- tax-incentive terms;
- project-specific infrastructure reimbursement mechanics;
- voluntary limits below otherwise applicable zoning thresholds; and
- project-specific representations concerning jobs, investment, tenant mix, or expansion.

If the City wants any of these concepts to become district-wide zoning rules, they should be rewritten to apply uniformly to all similarly situated M-2 businesses and reviewed for statutory authority.

State-law definitions reference (research, not codified): Iowa statutory and administrative definitions, plus proposed local definitions. The referenced glossary identifies the law controlling each quoted or paraphrased state definition and distinguishes it from municipal land-use terminology. State statutory definitions apply only within their respective contexts unless expressly incorporated in this ordinance.

H. Required Legal Review Before Introduction

The City Attorney should specifically review:

1. Whether the Major Industrial Facility thresholds and Special Exception structure fit Cascade's current Section 165.34 procedure;
2. Application of new performance standards to existing lawful uses and nonconformities;
3. Authority and process for applicant-funded independent studies and compliance testing;
4. Financial-assurance authority and acceptable decommissioning instruments;
5. Cost-recovery and infrastructure-agreement authority;
6. Public-record treatment of security-sensitive facility information;
7. State or federal preemption affecting utilities, air emissions, fuel storage, batteries, and emergency systems;
8. Whether any separate general noise, nuisance, fire-code, or building-code amendments should accompany the zoning amendment; and
9. Final numbering and integration with all amendments to Chapter 165 adopted after the base zoning text;

10. Whether an annual March 1 certification, third-party review deposit, generator documentation, commissioning deadline, and recorded decommissioning obligations require separate enabling ordinances, adopted schedules, or development agreements;
11. How Iowa Code §414.23 limits municipal zoning beyond Cascade city limits when county zoning applies, and the distinct plat-review authority under Iowa Code §354.9;
12. Whether a later separate engineering study supports measurable vibration standards; Jackson County's proposed peak-particle-velocity figures are not imported without independent review; and
13. Whether the requested generator equipment schedules or security-sensitive response plans require protection under Iowa Code Chapter 22 while preserving lawful public access; and
14. Whether interior-receptor levels, microphone placement, closed-window conditions, indoor access limitations, and modeled-versus-measured verification adequately protect occupied structures and can be administered fairly.

I. October 26 Advisory Packet and Community Suggestions

The 90-page October 27 Cascade Data Center Advisory Committee packet supplies new policy comparisons, primarily the Greene County and Decorah examples and more detailed provisions from a Jackson County **working draft**. It also reproduces the original October 6 Cascade M-2 discussion draft on PDF pages 58–72 and the October 1 Simple Mining presentation on pages 73–90. Reproduction in a packet does not make a proposed ordinance adopted law.

A separate submitted Word note asks about emissions and noise from approximately fifteen generators and a possible data center outside city limits but within two miles. These concerns are documented in the packet-review record and source transcription. They are addressed through equipment-specific documentation, noise compliance modeling, and jurisdictional research rather than a City attempt to set independent air-emissions limits or to regulate county-zoned parcels.

The Jackson County material supports considering annual reporting, independently reviewed acoustic testing, appropriate physical vibration measurement, objective applicant-funded review, a road-use agreement, and funded decommissioning. It does **not** establish those provisions as enacted Jackson County law or as scientifically validated thresholds for Cascade.

- Packet page/provenance index
- Community suggestions / source transcription
- Iowa Code §414.23: https://www.legis.iowa.gov/docs/code/2026/414.23.pdf
- Iowa Code §354.9: https://www.legis.iowa.gov/docs/code/2026/354.9.pdf

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1. Iowa General Assembly, *Iowa Code* §414.1 (2026), municipal regulation of building location and industrial land use, https://www.legis.iowa.gov/docs/code/2026/414.1.pdf. ↩
2. Iowa General Assembly, *Iowa Code* §414.2 (2026), uniformity for each class or kind of building throughout each zoning district, https://www.legis.iowa.gov/docs/code/2026/414.2.pdf. ↩↩
3. Iowa General Assembly, *Iowa Code* §414.3 (2026), comprehensive-plan and health, safety, conservation, and public-facility considerations, https://www.legis.iowa.gov/docs/code/2026/414.3.pdf. ↩

4. Iowa General Assembly, *Iowa Code* §414.12(2)-(4) (2026), Board of Adjustment special exceptions and variances, including 2025-added subsection (4), https://www.legis.iowa.gov/docs/code/2026/414.12.pdf. ↩↩

5. Iowa General Assembly, *Iowa Code* §423.3(95)(f) (2026), definitions applicable to data-center tax treatment, https://www.legis.iowa.gov/docs/code/2026/423.3.pdf. ↩

6. Iowa General Assembly, *Iowa Code* §103A.3 (2026), construction and building definitions, https://www.legis.iowa.gov/docs/code/2026/103A.3.pdf; City of Cascade, *Chapter 165 Zoning Ordinance*, §165.41, transcription in this repository: https://github.com/ttbakiatwoam/cascade-data-center/blob/master/dataset/cascade-zoning/CHAPTER-165-ZONING-ORDINANCE.md. ↩↩

7. Iowa General Assembly, *Iowa Code* §455B.261 (2026), water allocation, beneficial use, nonregulated use, and regulated use, https://www.legis.iowa.gov/docs/code/2026/455B.261.pdf. ↩↩↩

8. Iowa Department of Natural Resources, *Iowa Administrative Code* 567—22.1, air-quality definitions and emergency generators, https://www.legis.iowa.gov/docs/iac/rule/02-25-2026.567.22.1.pdf. ↩↩↩↩

9. Iowa General Assembly, *Iowa Code* §657.1 (2026), nuisance, https://www.legis.iowa.gov/docs/code/2026/657.1.pdf. ↩↩

10. Acoustical Society of America / American National Standards Institute, *ANSI/ASA S12.9*, environmental sound measurement and assessment standards, https://webstore.ansi.org/ (exact current edition and applicable part require acoustical expert verification). ↩↩↩↩

11. Iowa General Assembly, *Iowa Code* §§455B.131, 455B.381, and 455B.411 (2026), air contaminants, hazardous conditions, and waste-management terms, https://www.legis.iowa.gov/docs/code/2026/455B.pdf. ↩↩

12. Iowa General Assembly, *Iowa Code* §364.3 (2026), limitations on municipal topsoil and stormwater runoff requirements, https://www.legis.iowa.gov/docs/code/2026/364.3.pdf. ↩↩

13. Iowa General Assembly, *Iowa Code* §455B.301(9) (2026), definition of financial assurance instrument for sanitary disposal projects, https://www.legis.iowa.gov/docs/code/2026/455B.301.pdf. ↩

14. Iowa Department of Natural Resources, *Iowa Administrative Code* 567—60.1 (2026), wastewater and incorporated stormwater definitions, https://www.legis.iowa.gov/docs/iac/rule/567.60.1.pdf. ↩

15. City of Cascade, *Chapter 165 Zoning Ordinance*, original scanned M-2 Heavy Industrial District, "Entire zoning code-1.pdf," source PDF pp. 45–49 (printed code pp. 34–38), preserved at https://github.com/ttbakiatwoam/cascade-data-center/blob/master/sources/cascade-zoning/Entire%20zoning%20code-1.pdf; source transcription at https://github.com/ttbakiatwoam/cascade-data-center/blob/master/dataset/cascade-zoning/CHAPTER-165-ZONING-ORDINANCE.md. The same scan labels its 2009 zero-lot-line proposal a **draft** (PDF p. 6). ↩

16. City of Cascade, *Ordinance 05-22*, adopted October 10, 2022, adds fitness centers as M-2 B(18), https://github.com/ttbakiatwoam/cascade-data-center/blob/master/dataset/cascade-zoning/ordinances/ORDINANCE-05-22.md; *Ordinance 19-24*, adopted September 9, 2024, amends §165.12(12), including M-2 fencing, https://github.com/ttbakiatwoam/cascade-data-center/blob/master/dataset/cascade-zoning/ordinances/ORDINANCE-19-24.md. ↩

17. Acoustical Society of America / ANSI, *ASA/ANSI S12.72-2015 (R2025), Procedure for Measuring the Ambient Noise Level in a Room*, https://webstore.ansi.org/standards/asa/asaansis12722015r2025; *ASA/ANSI S12.9-2016/Part 7 (R2025), Measurement of Low-frequency Noise and Infrasound Outdoors in the Presence of Wind and Indoors in Occupied Spaces*, https://webstore.ansi.org/standards/asa/asaansis122016partr2025. Both are measurement frameworks; neither establishes the proposed numeric Cascade limits. ↩

18. International Organization for Standardization, *ISO 1996-2:2017, Acoustics—Description, Measurement and Assessment of Environmental Noise—Part 2*, https://www.iso.org/standard/59766.html. Contains guidance on indoor environmental-noise measurement; the draft's five-foot proximity criterion is a proposed local choice. ↩↩

19. World Health Organization, *Guidelines for Community Noise* (1999), indoor-bedroom night-noise guidance of 30 dBA Leq, https://www.who.int/publications-detail-redirect/a68672. This guideline is not binding Iowa law. ↩

20. U.S. Department of Energy, *Building Energy Codes Program, Iowa*, https://www.energycodes.gov/status/states/iowa; Iowa Administrative Code 481—301.24, *Residential Energy Code*, https://www.legis.iowa.gov/DOCS/ACO/IAC/LINC/Rule.481.301.24.pdf; Pacific Northwest National Laboratory, *Iowa Amendments to the 2012 IECC*,

https://www.pnnl.gov/main/publications/external/technical_reports/PNNL-24946Rev1.pdf. These cover energy, not building acoustics. ↩

21. Federal Highway Administration, *Noise Measurement Handbook*, §6.0, Table 6-1, FHWA-HEP-18-065 (2018), https://www.fhwa.dot.gov/Environment/noise/measurement/handbook.cfm. Typical building attenuation factors address highway traffic noise and are not site-specific industrial acoustic results. ↩↩

22. City of Cascade, *Code of Ordinances*, §6-12-8, "Building Code Adopted," Ordinance 02-22 (August 22, 2022), https://codelibrary.amlegal.com/codes/cascadeia/latest/cascade_ia/0-0-0-4159; U.S. Department of Energy, *Guide to Determining Climate Regions by County* (2015), https://www.energy.gov/eere/buildings/articles/building-america-best-practices-series-volume-73-guide-determining-climate. ↩

23. Federal Highway Administration, *Noise Measurement Handbook*, section 6 (building outdoor-to-indoor noise reduction), https://www.fhwa.dot.gov/Environment/noise/measurement/handbook.cfm; ASTM International, *ASTM E966*, facade field sound attenuation, https://store.astm.org/standards/e966; Iowa Supreme Court, *Singer v. City of Orange City*, No. 23-1600 (Dec. 20, 2024), https://www.iowacourts.gov/iowa-courts/supreme-court/supreme-court-opinions/case/23-1600. The City Attorney should review access and enforcement authority. ↩

24. Federal Highway Administration, *Construction Noise Handbook*, §5.3–5.4 (baseline measurement and comparable before/after locations and conditions), https://www.fhwa.dot.gov/Environment/noise/construction_noise/handbook/handbook05.cfm; *Noise Measurement Handbook*, discussion of measurement comparability and uncertainty, https://www.fhwa.dot.gov/Environment/noise/measurement/handbook.cfm; Iowa General Assembly, *Iowa Code* §657.1 (2026), nuisance and civil remedies, https://www.legis.iowa.gov/docs/code/2026/657.1.pdf. This draft evidence standard is a proposed local ordinance provision, not an established precondition to an Iowa-law nuisance claim. ↩↩

25. ASHRAE, *ANSI/ASHRAE Standard 55-2023, Thermal Environmental Conditions for Human Occupancy*, https://www.ashrae.org/technical-resources/bookstore/standard-55-thermal-environmental-conditions-for-human-occupancy; ASHRAE, *Technical FAQs*, "recommended indoor temperature and humidity levels for homes," https://www.ashrae.org/technical-resources/technical-faqs (comfort varies by humidity, clothing, activity, and season, so no universal 65–75°F required range); ASHRAE, *2023 Handbook—HVAC Applications*, ch. 49, "Noise and Vibration Control," https://handbook.ashrae.org/Handbooks/A23/SI/A23_Ch49/a23_ch49_si.aspx (document actual HVAC operating conditions and distinguish equipment from background sound). ↩